

Decision Tree Document for Selecting RDBMS Technology

Introduction

This document is designed to help product teams within our enterprise make informed decisions when choosing the appropriate RDBMS technology for new Greenfield development projects. The decision tree takes into account our existing infrastructure, technical requirements, and enterprise-specific needs.

Assumption is a need for the data store that has a strong ACID compliance requirement.

Key Considerations

- 1. Existing Technologies Supported:**
 - a. Oracle
 - b. Oracle on Exadata
 - c. SQL Server
- 2. Development Language:**
 - a. Java
- 3. Data Replication:**
 - a. Oracle GoldenGate
- 4. Team Expertise:**
 - a. Strong proficiency in Oracle technologies
- 5. Industry Requirements:**
 - a. Financial sector compliance and audit requirements
- 6. Production Environment Requirements:**
 - a. Disaster Recovery (DR)
 - b. High Availability (HA)
- 7. Security Considerations:**
 - a. Minimize exposure to CVE vulnerabilities
- 8. Reporting Capability:**
- 9. High Availability**

Decision Tree Questions

- 1. Performance Requirements:**

- a. Do you need high-performance computing for large-scale transaction processing and analytics?
 - b. Are you dealing with smaller queries that require efficient handling?
- 2. Scalability Needs:**
 - a. Does your project require a highly scalable solution that can handle growing data volumes and workloads?
 - b. Is scalability less of a concern for your current and near-future needs?
- 3. Budget Constraints:**
 - a. What is your budget for licensing, hardware, and maintenance?
 - b. Are you looking for a cost-effective solution with lower upfront costs?
- 4. Platform Compatibility:**
 - a. Are you using Windows, Linux, Unix, or a combination of these platforms?
 - b. Does your team have expertise in managing databases on specific platforms?
- 5. Advanced Features:**
 - a. Do you need advanced enterprise capabilities like distributed databases and failover?
 - b. Are you looking for built-in business intelligence and analytics tools?
- 6. Maintenance and Management:**
 - a. Does your team have the expertise to manage and maintain Oracle databases?
 - b. Are you looking for automated database tools leveraging machine learning for easier management?
- 7. Integration Needs:**
 - a. Do you need seamless integration with other Oracle products or third-party applications?
 - b. Are you looking for simpler data replication across systems?
- 8. Security Requirements:**
 - a. How critical is minimizing CVE vulnerabilities for your project?
 - b. What security features are essential for compliance and audit requirements?
- 9. Disaster Recovery and High Availability:**
 - a. What are your DR and HA requirements for the production environment?
 - b. Do you need built-in support for DR and HA?

Decision Tree Structure

- 1. Performance Requirements:**
 - a. High Performance → Oracle on Exadata

- i. Less than 0.1 ms response time for single block reads.
- b. Efficient Handling of Smaller Queries → SQL Server
- 2. Scalability Needs:**
 - a. High Scalability → Oracle on Exadata
 - i. Compute/ Storage can be horizontally added.
 - b. Lower Scalability Needs → Oracle Database or SQL Server
- 3. Budget Constraints:**
 - a. Higher Budget → Oracle on Exadata
 - b. Lower Budget → SQL Server
- 4. Platform Compatibility:**
 - a. Windows, Linux, Unix → Oracle
 - b. Windows, Linux → SQL Server
- 5. Advanced Features:**
 - a. Advanced Enterprise Capabilities → Oracle on Exadata
 - i. Compression/ In built columnar data representation, Smart scan for OLAP, MPP storage servers.
 - b. Built-in Business Intelligence and Analytics → SQL Server
- 6. Maintenance and Management:**
 - a. Expertise in Oracle Management → Oracle
 - b. Automated Tools for Easier Management → SQL Server
- 7. Integration Needs:**
 - a. Integration with Oracle Products → Oracle on Exadata
 - i. ODI, Goldengate
 - b. Simpler Data Replication → SQL Server
- 8. Security Requirements:**
 - a. Minimize CVE Vulnerabilities → Oracle on Exadata
 - i. Quarterly patches available, All Oracle db. systems are always in n-1 patch set.
 - b. Adequate Security with Lower Cost → SQL Server
- 9. Disaster Recovery and High Availability:**
 - a. Robust DR and HA Support → Oracle on Exadata
 - i. GDS, RAC, Local Standby support
 - ii. Flashback technology to support end of day snapshot
 - b. Adequate DR and HA Support → SQL Server

Technical Comparison Matrix

Technical Aspect	Oracle	Oracle on Exadata	SQL Server
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Performance	High Performance	Optimized for High Performance	Excellent for OLTP and OLAP
Scalability	Scalable	Highly Scalable	Scalable with Big Data Clusters
Cost	Significant	Higher Investment	More Cost-Effective
Licensing	Per Core or NUP	Per Core or NUP	Per Core or Server+CAL
Platform Support	Linux	Linux	Windows
Advanced Features	RAC, Data Guard, Security, GDS, Read only Standby for Reporting	RAC, Data Guard, Security, GDS, Read only Standby for Reporting	BI Tools, ML Services, Always On
Maintenance	Skilled DBAs Required	Exadata Management Tools	Automated Maintenance Tools
Integration	Seamless with Oracle Products	Seamless with Oracle Products	Strong with Microsoft Ecosystem
Security	Advanced Security Features	Minimized CVE Vulnerabilities	Adequate Security
DR and HA	Robust DR and HA Support	Built-in DR and HA Capabilities	Adequate DR and HA Support

Conclusion

By following this decision tree and considering the technical comparison matrix, product teams within the enterprise can make informed decisions on selecting the appropriate RDBMS technology for new Greenfield development projects. This approach ensures alignment with technical requirements, team expertise, and industry-specific needs.